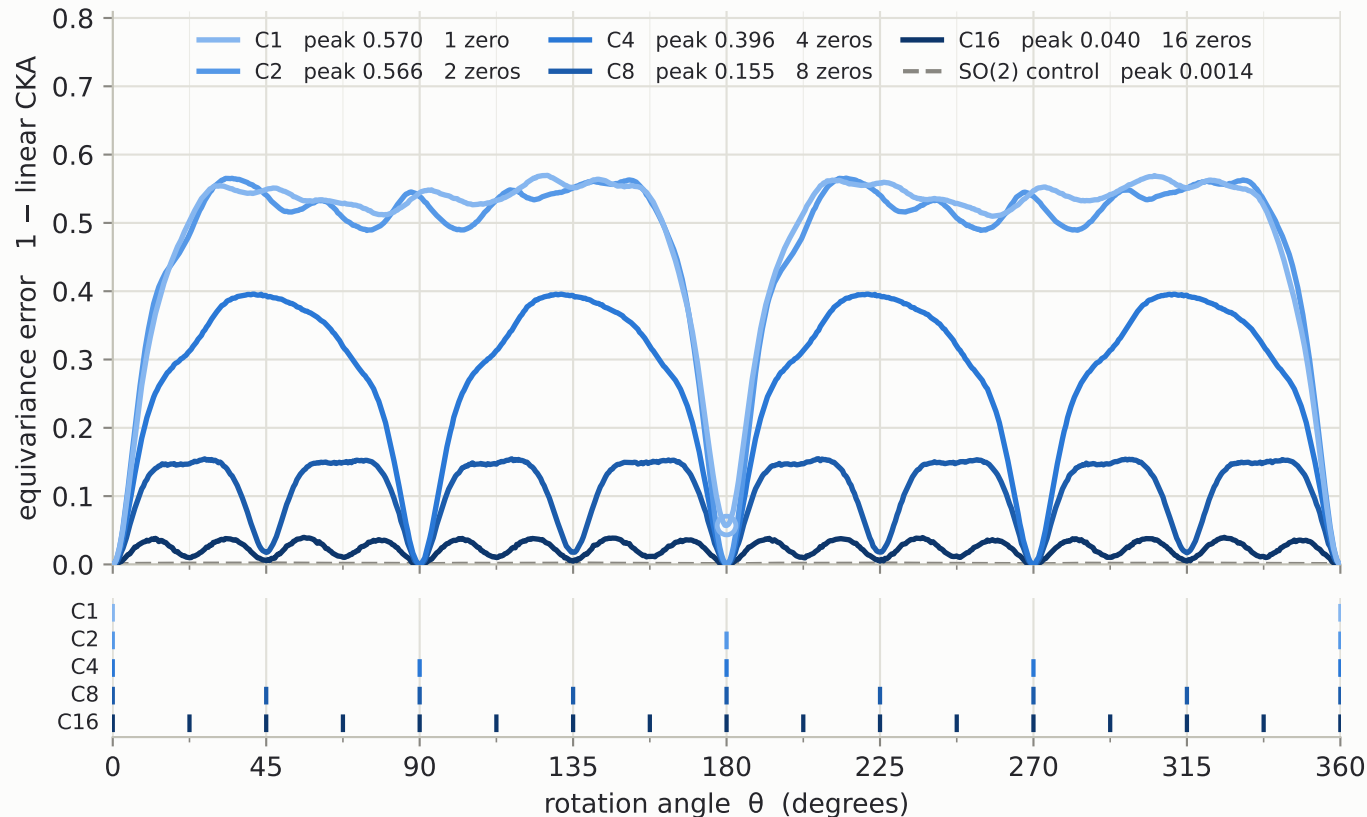
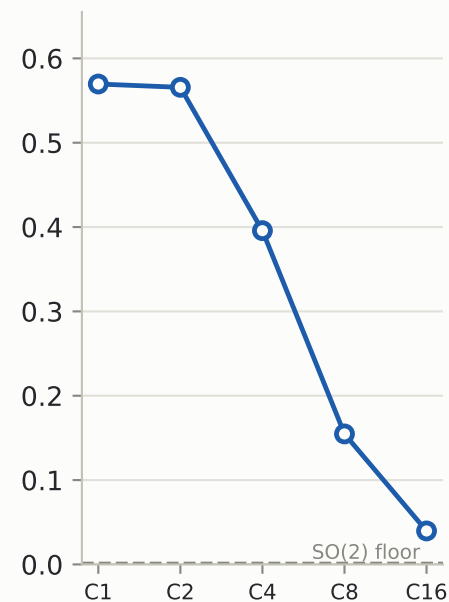


CKA recovers the rotation group without ever being told $\rho(g)$

C_N -equivariant escnn models, untrained; mean of 5 seeds; error touches 0 at every multiple of $360/N$



peak error vs. group order



○ C1 has no group, yet its error notches at 180° — down to 10% of peak, against 96% at 90° . Any rotation-invariant readout is dominated by low angular harmonics, and a 180° rotation maps harmonic m to $(-1)^m$ — every even one survives. The notch appears in all four readouts tried (rings, disk-mean, un-rotation, harmonic-magnitude), so it reflects invariant readouts plus the square lattice, not this metric.